

Consumer-Aligned Learning for Quality-of-Transmission: The False-Clear Rate at the FEC Cliff

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Abstract: An ML-QoT estimator trained on average GSNR error is misaligned with the FEC decoder’s threshold: it false-clears at the cliff. On GNPY over CORONET-CONUS a consumer-aware objective cuts false-clears $\sim 5\times$ at equal capacity.

1 Introduction

Machine-learning Quality-of-Transmission (ML-QoT) estimation is a leading route to reducing the system margin—the deliberately stranded capacity—that optical operators carry against uncertain reach performance [2, 3]. ML-QoT models are almost always trained to minimise a *prediction error* (mean-squared or mean-absolute GSNR/OSNR error). We argue, and measure, that this objective is misaligned with the consumer it serves. The consumer of a QoT estimate is the *format-selection decision*, and ultimately the forward-error-correction (FEC) decoder, which reads a *threshold*: a lightpath either meets its format’s required GSNR or it does not. An estimator can be excellent on average yet, near a format boundary, over-predict just enough to select a rate the link cannot support—a *false-clear* (an in-service outage), even though its mean error is small.

We make the false-clear rate a first-class, measured KPI and show three things on the GN-model physical layer of GNPY [1] over the CORONET-CONUS reference network: (i) an average-error ML-QoT objective false-clears at the FEC cliff, while a consumer-aware objective does not, at equal capacity; (ii) the underlying QoT margin is *consumer-relative*—which lightpaths are at risk is a footprint property (spectral position and reach), so no blanket margin fits all; and (iii) a fill-driven *refresh floor* sets when a certificate must be re-validated.

2 Method

Substrate. Lightpaths are transparent shortest-path routes of the **CORONET-CONUS** reference topology (75 nodes; ≤ 2500 km, $2\text{--}31\times 80$ km spans). Per-channel GSNR is computed with the GN model in GNPY [1] on standard single-mode fibre (0.2 dB/km, $D = 17$ ps/nm/km), EDFAs (NF 6 dB), +2 dBm/ch, 32 GBd, 50 GHz grid, from a lightly filled “provisioning” band (6 channels) to a full C-band (76 channels).

Certificate, consumer, witness. A certificate selects the highest format whose required GSNR the estimate clears.

The *consumer* is that decision; the *witness* is the true deployed GSNR (the receiver’s pre-FEC BER). For the ML study, features are provisioning-time GSNR and (spectral position, reach, fill), the label carries an irreducible unmeasured-state term, and two gradient-boosted objectives are compared: *average-error* (squared loss) and *consumer-aware* (a conservative lower-quantile loss that protects the decision, not the mean). Bars and manipulation checks are pre-registered; the sealed record is content-hashed with disjoint graded seeds [5].

3 Results

3.1 ML-QoT trained on the wrong metric

Under irreducible prediction uncertainty, the average-error estimator attains lower mean error yet false-clears far more at the FEC cliff than the consumer-aware objective: MAE 0.65 vs 0.95 dB (it wins reconstruction) but false-clear ≈ 0.041 vs ≈ 0.008 ($\sim 5\times$) at comparable delivered spectral efficiency (≈ 3.9 bits/symbol). Figure 1(a) shows the reconstruction-vs-consumer dissociation; (b) shows that the average-error false-clears concentrate where the GSNR headroom above the nearest format threshold is small—the cliff—exactly where a symmetric error metric is blind. The prescription is to train ML-QoT on the consumer’s threshold decision, not on volume-averaged fidelity.

3.2 The margin is consumer-relative

The same threshold-blindness afflicts the margin itself under incremental fill. As a lightpath’s band fills toward its deployed load, its GSNR drops, and whether it crosses a format threshold depends on its *footprint*: false-clears concentrate on the services whose spectral position and reach place their GSNR nearest a boundary, and are absent elsewhere (Fig. 2). Consequently a single blanket margin cannot be right for all. We quantify the cost against today’s practice—one *uniform* system margin sized so the fleet stays safe—using *real* deployed fill: each lightpath’s channel load is drawn from the per-link

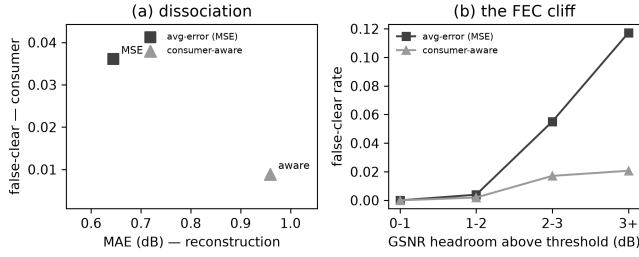


Figure 1: ML-QoT reconstruction-vs-consumer dissociation (GNPy + gradient-boosted regressors): (a) the average-error objective attains lower MAE yet higher false-clear than the consumer-aware objective; (b) it false-clears at the FEC cliff (small GSNR headroom), where the consumer-aware objective does not. Marker shape distinguishes objectives.

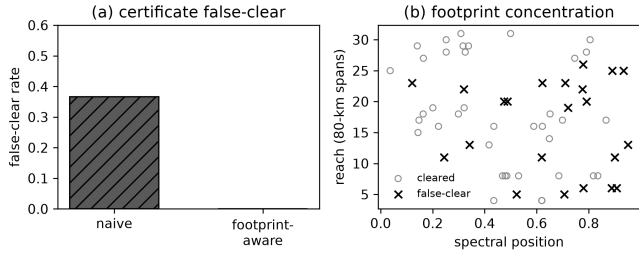


Figure 2: Consumer-relativity of the QoT margin on GNPy over CORONET-CONUS: per-service false-clear over (spectral position, reach) concentrates on threshold-adjacent services; a footprint-aware certificate removes it. Marker shape distinguishes classes.

fill distribution of the SNDlib **janos-us-ca** demand matrix (61 links; median 10, 90th-percentile 35 of 76 channels [4]). Real traffic is heavy-tailed, so a uniform margin sized for the few near-full links over-provisions the lightly-filled majority. Giving *both* policies the same QoT-estimator uncertainty (0.5 dB RMSE, so the footprint-aware certificate is not assumed perfect), a certificate that sizes each margin to the lightpath’s own fill recovers $\approx 8\%$ of spectral efficiency (≈ 0.33 b/sym, ~ 21 Gb/s per channel) at matched safety (false-clear ≤ 0.01), on a rate-flexible transceiver model. The gain is modest but free: it is capacity the uniform margin strands on the majority of links that never fill.

3.3 The refresh floor

Because the risk is driven by the fill *increment* between provisioning and use, a certificate has a refresh horizon. Sweeping the deployed fill from 6 to 76 channels, the naive false-clear rises approximately linearly with the number of added channels ($FC \approx 0.008\Delta N$, $R^2 = 0.95$); at a 0.05 target the refresh floor is ≈ 8 added channels. The rule is concrete: re-certify a service before the band around it grows by ~ 8 channels, not on a fixed calendar.

4 Related work

QoT estimation and physical-layer-aware routing/spectrum assignment are mature [1, 3]; ML-QoT is active, with estimators trained predominantly on prediction error [2]. Our contribution is orthogonal: not a better estimator or allocator, but a *measurement*—the false-clear rate as a first-class, consumer-relative KPI—and the finding that the ML-QoT training *objective*, and the margin allocation, should target the consumer’s threshold.

5 Discussion

Train QoT models, and size QoT margin, for the decision the FEC decoder makes, not for average fidelity. The GN model is the evidence rung; a coherent-receiver pre-FEC-BER witness on a fibre testbed, and the EGN/split-step model, are the external-validity graduations. (The consumer-relative, false-clear methodology is not optics-specific [5], but the results here stand on the optical QoT certificate alone.)

References

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